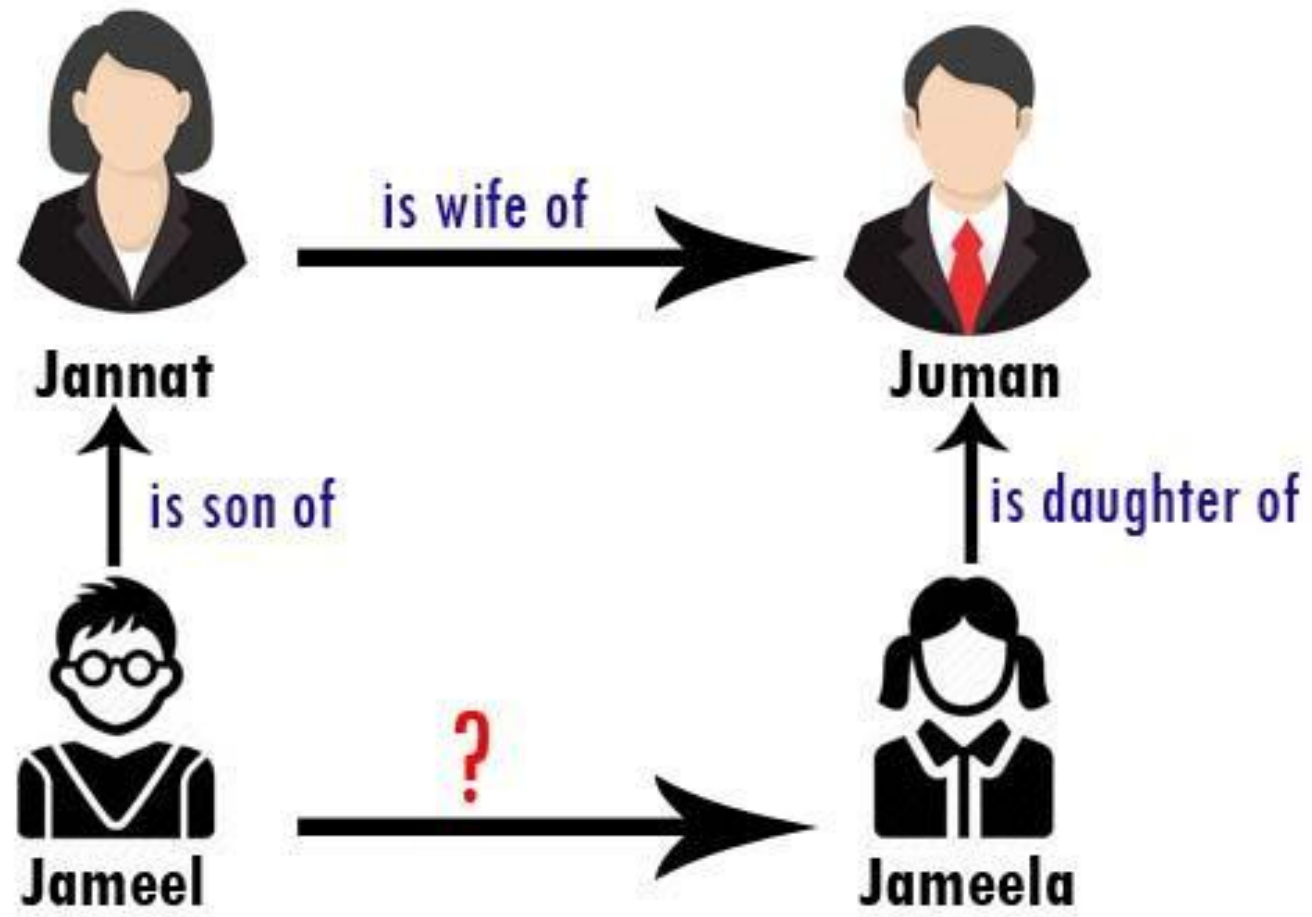


Semantic Networks

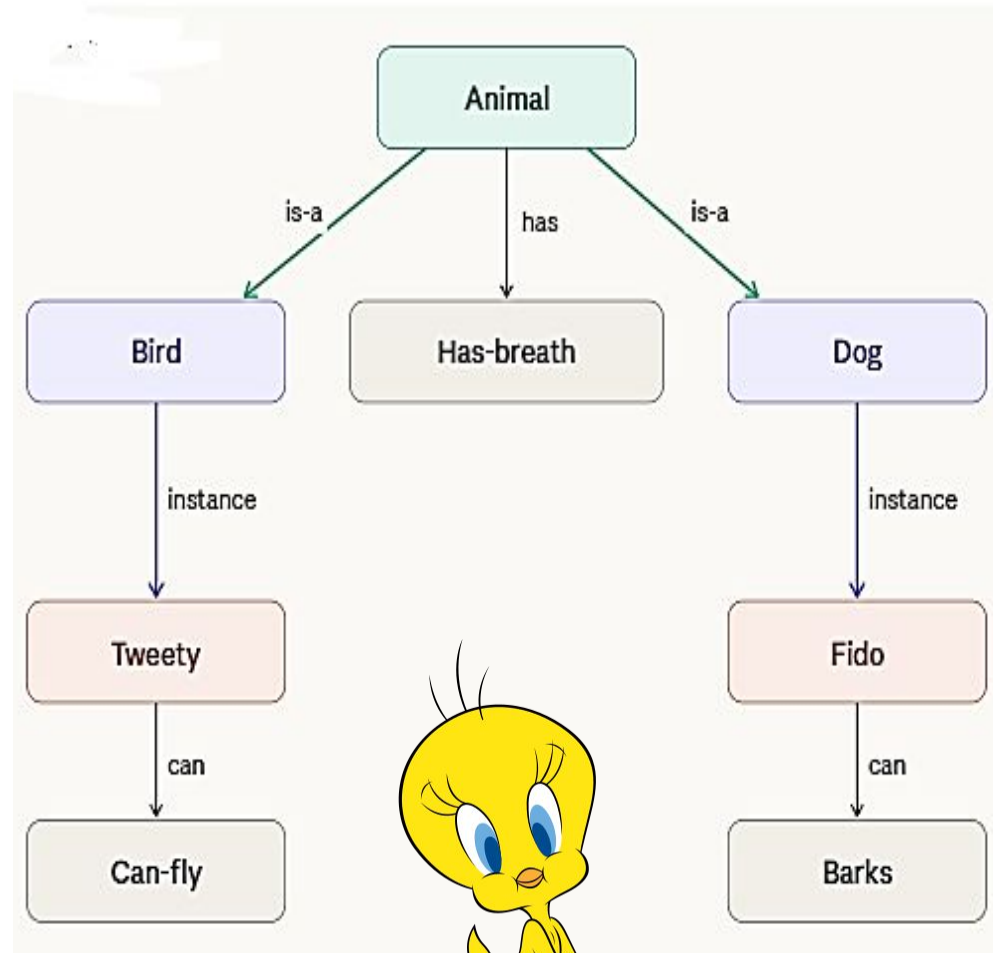
In Artificial Intelligence

Remember?

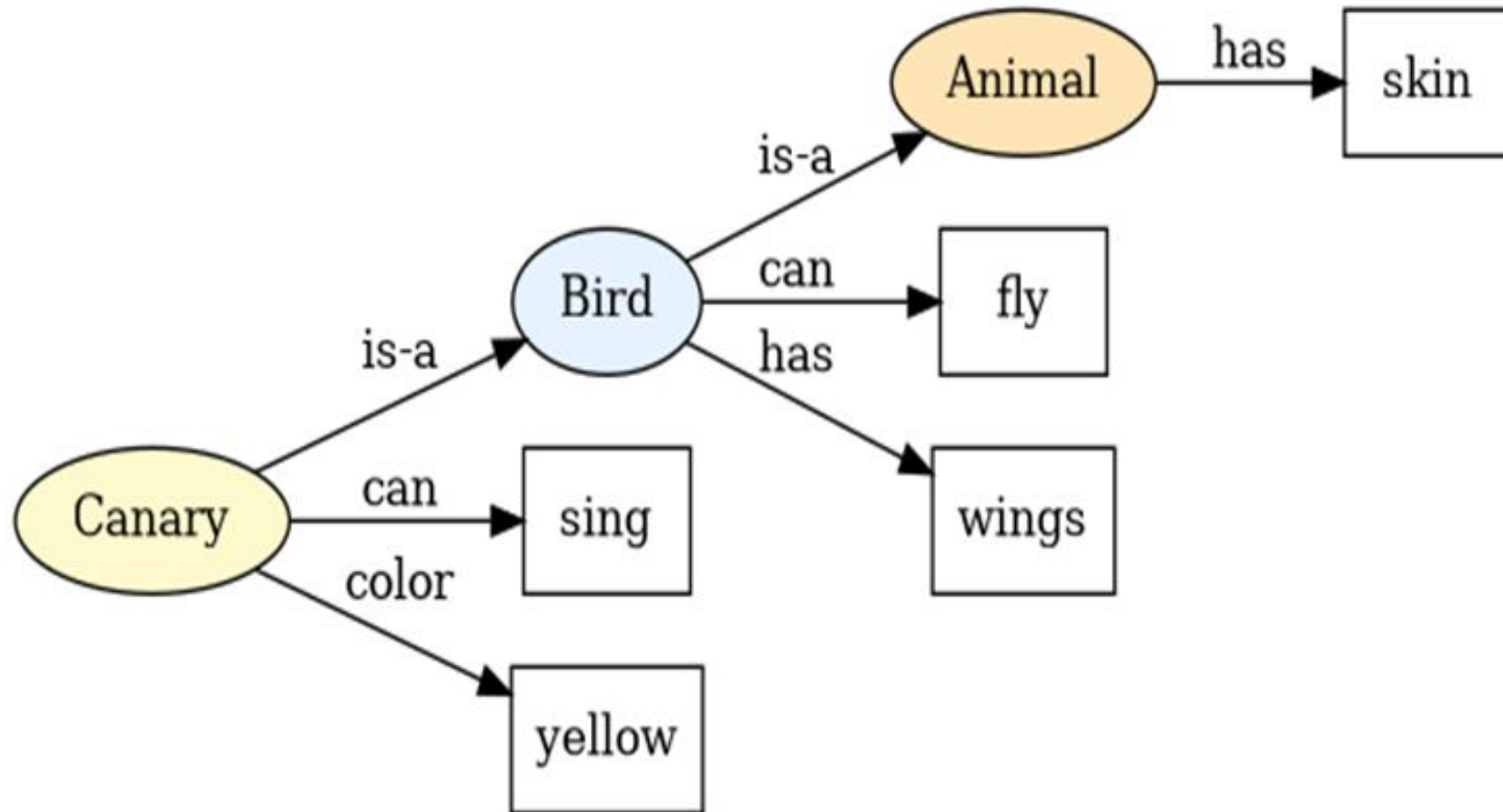


Semantic Networks

- ▶ A Semantic Network is a way to represent **knowledge as a graph**
- ▶ Represent and organize knowledge in a way that machines can understand.
- ▶ These networks connect different concepts (nodes) through relationships (edges) such as "is a", "part of" or "related to" allowing AI systems to make sense of how ideas are linked.
- ▶ Its not about just storing facts, but storing how facts relate to each other.



Semantic Networks

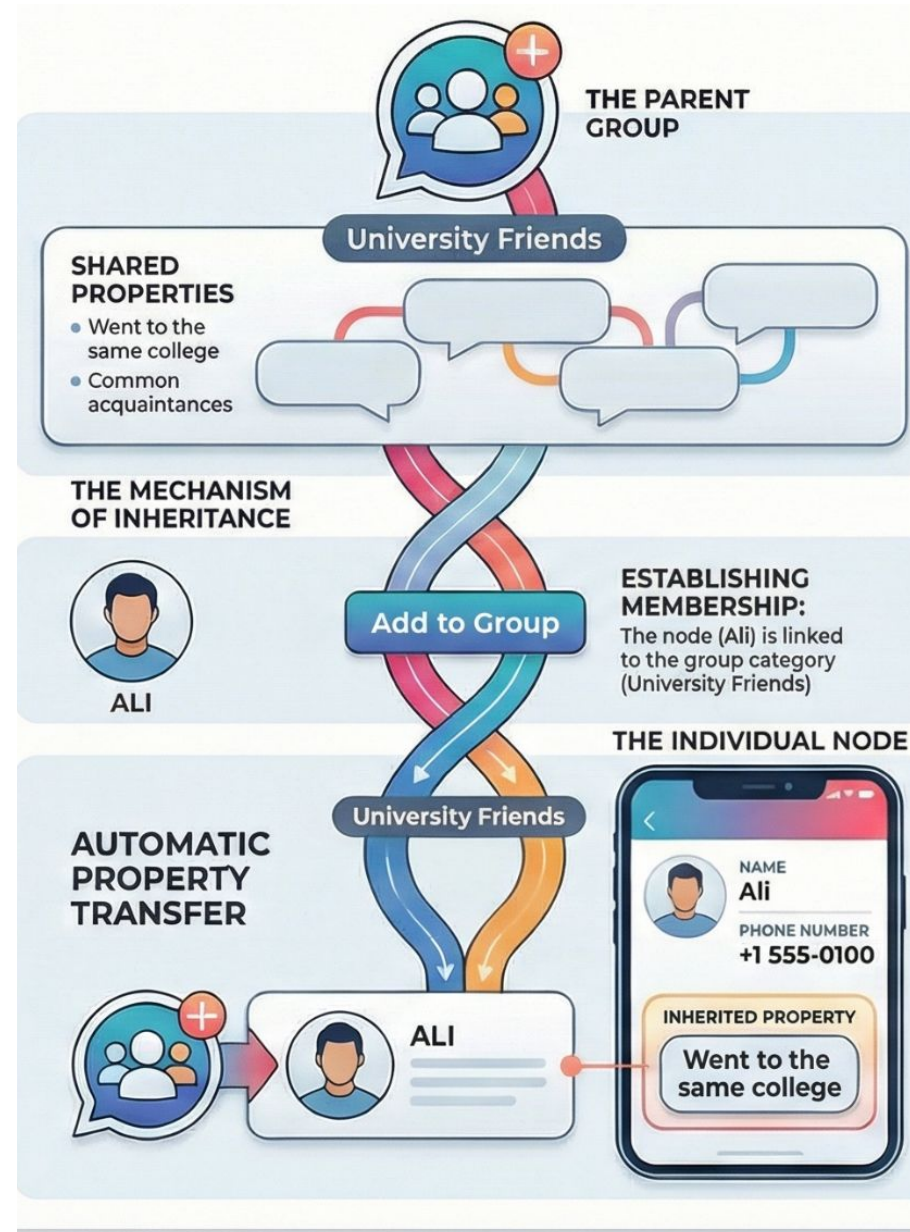


Semantic Networks

- ▶ Core Concepts:
 - ▶ **Nodes:** represent concepts, objects, or instances (Animal, Bird, Tweety)
 - ▶ **Edges (links):** represent relationships.
 - ▶ is-a → inheritance (Bird is-a Animal)
 - ▶ instance-of → a specific object (Tweety is an instance of Bird)
 - ▶ has, can, part-of → property relationships
 - ▶ **Inheritance:** If Animal "has breath," then Bird inherits that property automatically. You don't have to state it again for every bird.
 - ▶ Tweety can breathe (nobody ever told the system that directly.)

Semantic Networks

- **Inheritance:** The concept/node will share all relevant properties of the domain.
- Think of it like WhatsApp contacts.
- Your contact 'Ali' has a phone number, a name, and belongs to a group called 'University Friends'. The group has shared properties, everyone in it went to the same college. Ali *inherits* that fact just by being a member."



MAPPING KNOWLEDGE: A QUICK GUIDE TO SEMANTIC RELATIONSHIPS

TAXONOMY & PROPERTIES

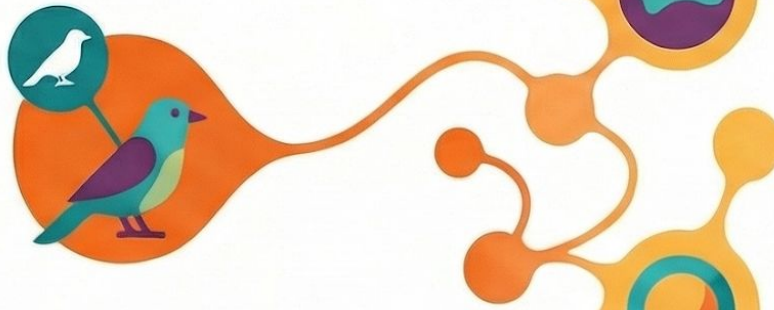
is-a (Subclass)

Connects a specific category to a broader, more general class.



instance-of (Specific Object)

Identifies an individual entity as a member of a specific class.



has-property (Attribute)

Assigns a characteristic or inherent quality to a defined object.



COMPOSITION & CAPABILITY

part-of (Composition)

Illustrates how individual components comprise a larger, complex whole.



can (Capability)

Defines the functional abilities or specific actions an entity can perform.



located-in (Spatial)

Specifies the geographic or physical placement of one object within another.



A network graph with nodes of various sizes and colors (green, blue, red, yellow) connected by gray lines. The text "KNOWLEDGE GRAPH" is centered in the middle.

**KNOWLEDGE
GRAPH**

What is a Knowledge Graph?

- ▶ A Knowledge Graph is a **large-scale, real-world semantic network** stored as a graph database, where entities and their relationships are enriched with **attributes, context, and reasoning capabilities**.
- ▶ Knowledge Graph = Entities + Relationships + Attributes + Context
- ▶ When you Google 'Cristiano Ronaldo', why does a panel appear on the right showing his age, nationality, club, wife, and children – without you clicking anything?
- ▶ Some URLs to see KGs:
 - ▶ query.wikidata.org
 - ▶ <https://kumu.io/>
 - ▶ <https://graphcommons.com/>

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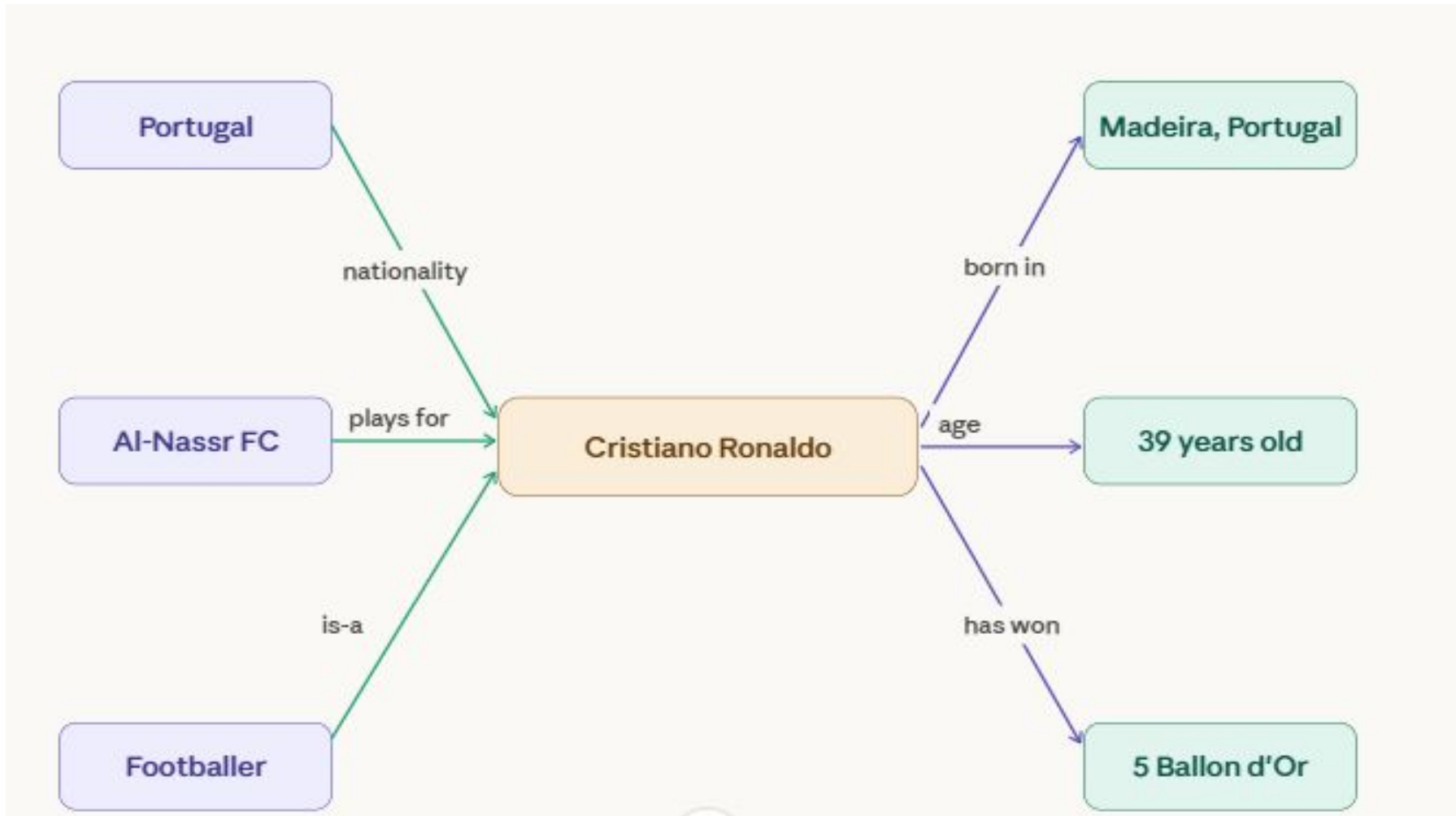


About

Cristiano Ronaldo dos Santos Aveiro, nicknamed CR7, is a Portuguese professional footballer who plays as a forward for and captains both Saudi Pro League club Al-Nassr and the Portugal national team. [Wikipedia](#)

Born	February 5, 1985 (age 41 years), Hospital Dr. Nélío Mendonça, Funchal, Portugal
Current teams	Al-Nassr FC (#7 / Forward), Portugal national football team (#7 / Forward)
Dates joined	2023 (Al-Nassr FC), 2021 (Manchester United F.C.) · See more
Children	Cristiano Ronaldo Jr, Angel dos Santos Aveiro
Salary	208.4 million EUR (2026)
Full name	Cristiano Ronaldo dos Santos Aveiro
Siblings	Hugo dos Santos Aveiro, Katia Aveiro, Elma dos Santos Aveiro

Knowledge Graph



Knowledge Graph: The Triple

- ▶ Everything in a Knowledge Graph is stored as a triple:
 - ▶ (Subject) → (Predicate) → (Object)
 - ▶ (Ronaldo) → (plays for) → (Al-Nassr)
 - ▶ (Al-Nassr) → (located in) → (Saudi Arabia)
 - ▶ (Saudi Arabia) → (is a) → (Country)
- ▶ This is also called RDF (Resource Description Framework), the web standard for Knowledge Graphs.
- ▶ A hospital Knowledge Graph might store:
 - ▶ (Patient_Ali) → (diagnosed with) → (Type 2 Diabetes)
 - ▶ (Type 2 Diabetes) → (treated by) → (Metformin)
 - ▶ (Metformin) → (interacts with) → (Ibuprofen)
 - ▶ (Patient_Ali) → (currently takes) → (Ibuprofen)
 - ▶ **ALERT! The system can *infer* a dangerous drug interaction automatically by traversing the graph – no doctor explicitly programmed that rule.**

Practice: Draw the Knowledge Graph

► People:

- Ahmed is a student at Aror University
- Ahmed is enrolled in the course "Artificial Intelligence"
- Ahmed is supervised by Dr. Faheem
- Dr. Faheem is a professor at Aror University
- Dr. Faheem teaches the course "Artificial Intelligence"
- Dr. Faheem has published a paper titled "Deep Learning for Hallucination Diagnosis"

► Courses & Department:

- "Artificial Intelligence" belongs to the AI Department
- AI Department is part of Aror University
- Aror University is located in Sukkur, Pakistan

► Paper:

- "Deep Learning for Hallucination Diagnosis" is published in the journal "IEEE Access"
- "Deep Learning for Hallucination Diagnosis" is of type Research Paper

► Tasks

- Identify the TriplesList all facts as
- Subject → Predicate → Object triples.
Example: (Ahmed) → (enrolled-in) → (Artificial Intelligence)
- Draw the Graph Using nodes and labeled edges

► Answer Following

- Which university does Dr. Faheem's student belong to?
- In which city is Ahmed studying?
- What type of document did Ahmed's supervisor publish?
- Which journal is connected to the AI Department (indirectly)?

daily specials

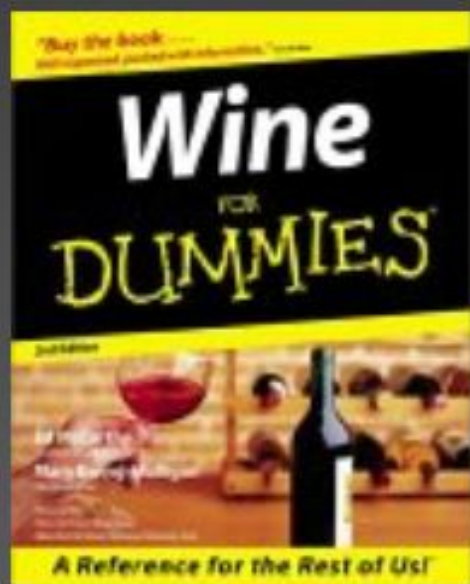


Which wine
should
I serve with
seafood
today?

**A shared
ONTOLOGY
of
wine and food**

**French wines
and
wine regions**

**California
wines and
wine regions**



Ontology (Greek *ont-* ("being") + *-logia* ("study of"))

- ▶ An ontology is an explicit formal description of a domain:
 - ▶ Concepts
 - ▶ properties and attributes of concepts
 - ▶ constraints on properties and attributes
 - ▶ Individuals (*often, but not always*)
- ▶ An ontology defines
 - ▶ a common vocabulary
 - ▶ a shared understanding
- ▶ **Philosophical definition:**
 - ▶ Ontology is the branch of philosophy studying the nature of being, existence, and reality, questioning what entities exist and how they are categorized.
 - ▶ It covers topics like metaphysical realism, which assumes an objective reality independent of observation

Ontology

- ▶ In computer science, it is a formal, structured framework defining concepts, relationships, and rules within a specific domain to enable knowledge sharing and AI reasoning.
- ▶ An ontology acts as a model (e.g., knowledge graphs) that structures data using classes, attributes, and relations to improve AI efficiency, data traceability, and understanding.
 - ▶ Taxonomies on the Web (<https://schema.org/>)
 - ▶ Catalogs for on-line shopping (Amazon)
 - ▶ Domain-specific standard terminology
 - ▶ Unified Medical Language System (UMLS)
 - ▶ United Nations Standard Products and Services Code (UNSPSC) - terminology for products and services

Ontology Engineering

- ▶ Defining terms in the domain and relations among them
 - ▶ Defining concepts in the domain (classes)
 - ▶ Arranging the concepts in a hierarchy (subclass-superclass hierarchy)
 - ▶ Defining which attributes and properties (slots) classes can have, and constraints on their values
 - ▶ Defining individuals and filling in slot values
- ▶ How ontology engineering can be beneficial?
 - ▶ To enable reuse of domain knowledge (to avoid “re-inventing the wheel”)
 - ▶ to introduce standards to allow interoperability

Ontology Engineering (*Building blocks*)

Classes

Categories of things

Properties

Relations & Attributes

Individuals

Specific instances

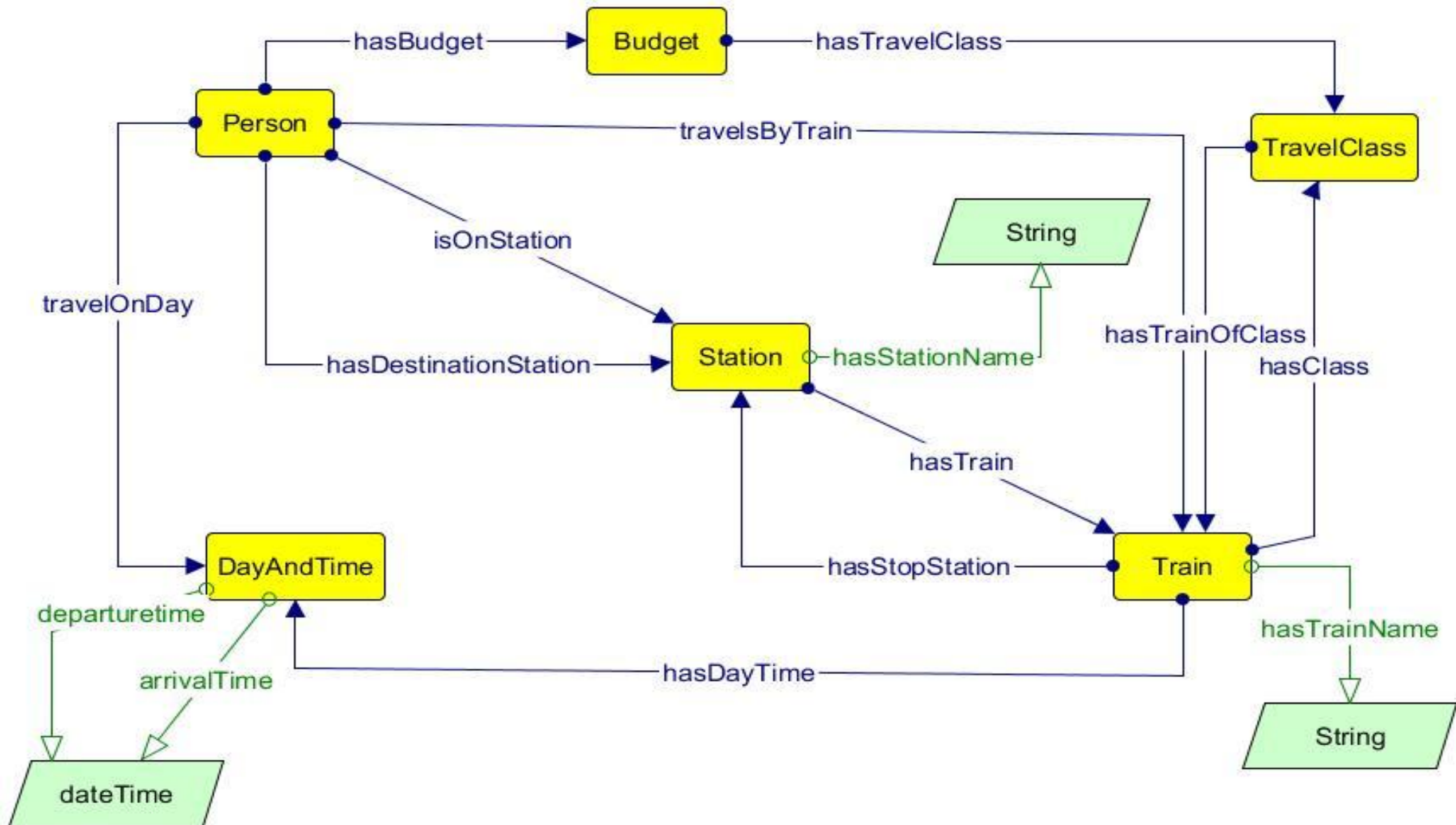
Axioms

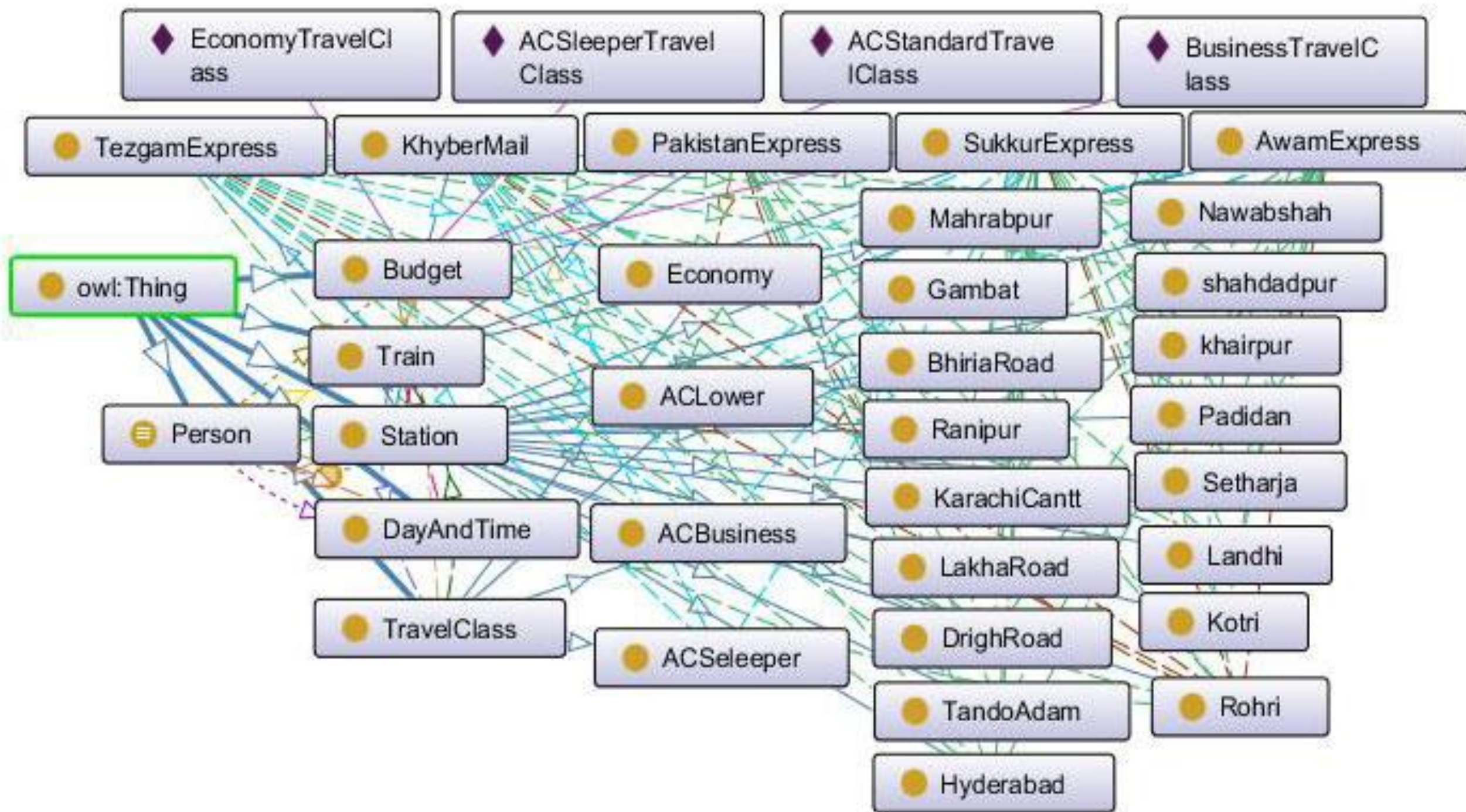
Rules and constraints

Ontology Engineering

- ▶ Design Principles:

- ▶ **Clarity:** Every class and property must have a precise, unambiguous definition using `rdfs:comment` or `skos:definition`. Avoid terms that mean different things to different people.
- ▶ **Coherence:** All axioms must be logically consistent. A reasoner should never infer a contradiction. For example, you can't have a class that is simultaneously a subclass of Drug AND Disease unless that's intentional.
- ▶ **Extensibility:** Design for reuse. Use established upper ontologies so others can integrate with your ontology easily.





RDF → RDFS → OWL

OWL (Web Ontology Language)

- Most Expressive
- Classes, restrictions, logic, reasoning
- OWL DL / OWL Full

RDFS (RDF Schema)

- Schema Layer
- Class hierarchy, property ranges
- Rdfs: subClassOf

RDF (Resource Description Framework)

- Data Layer
- Triples (Subject, Predicate, Object)
- Rdf:type

RDF (Resource Description Framework)

- ▶ every statement is broken into three parts.
 - ▶ <Subject> <Predicate> <Object>
 - ▶ Everything (subjects, predicates, objects) is identified by a **URI** (Uniform Resource Identifier: a web address). This makes knowledge globally unambiguous.
 - ▶ Turtle syntax (most readable RDF format)

```
@prefix : <http://med.example.org/> .  
@prefix rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#> .  
:Aspirin :treats :Headache .  
:Aspirin :hasFormula "C9H8O4" .  
:Ali :hasDiagnosis :Headache .
```

- ▶ Each line is one triple.
- ▶ rdf:type defines the class

```
:Aspirin rdf:type :Drug .  
:Ali rdf:type :Patient .
```

RDFS: The Schema Layer

- ▶ RDF alone has no concept of class hierarchy or property constraints.
- ▶ RDFS adds that vocabulary.
- ▶ The two most important keywords:
 - ▶ `rdfs:subClassOf`, establishes inheritance between classes.
 - ▶ `rdfs:domain` / `rdfs:range`, constrains what a property connects.

```
@prefix rdfs: <http://www.w3.org/2000/01/rdf-schema#> .  
# Class hierarchy  
:NSAID      rdfs:subClassOf :Drug .  
:Aspirin    rdfs:subClassOf :NSAID .  
# Property constraints  
:treats     rdfs:domain  :Drug ;  
            rdfs:range   :Disease .  
:hasDiagnosis rdfs:domain :Patient ;  
            rdfs:range   :Disease .  
# Human-readable labels  
:Drug       rdfs:label   "Drug"@en ;  
            rdfs:comment "A chemical substance used as medication."@en .
```


OWL (Web Ontology Language)

- ▶ OWL is where ontologies become truly powerful.
- ▶ It adds Description Logic on top of RDFS, enabling a *reasoner* to check consistency and infer new facts automatically.

Feature	OWL construct	Meaning
Class equivalence	owl:equivalentClass	Two names, same class
Disjoint classes	owl:disjointWith	No overlap allowed
Property inverse	owl:inverseOf	treats $\leftrightarrow$ isTreatedBy
Cardinality	owl:maxCardinality	A patient has ≤ 1 primary doctor
Existential restriction	owl:someValuesFrom	Every drug treats at least one disease
Universal restriction	owl:allValuesFrom	All diagnoses are diseases
Transitive property	owl:TransitiveProperty	isPartOf chains

OWL (Web Ontology Language)

```
@prefix owl: <http://www.w3.org/2002/07/owl#> .

# Class declarations
:Drug    rdf:type owl:Class .
:Disease rdf:type owl:Class .
:Patient rdf:type owl:Class .

# Disjoint: nothing can be both a Drug and a Disease
:Drug owl:disjointWith :Disease .

# Existential restriction: every Drug must treat at least one Disease
:Drug rdfs:subClassOf [
  rdf:type      owl:Restriction ;
  owl:onProperty :treats ;
  owl:someValuesFrom :Disease
] .

# Inverse property
:isTreatedBy rdf:type      owl:ObjectProperty ;
              owl:inverseOf :treats .

# Cardinality: a patient has at most 1 primary physician
:Patient rdfs:subClassOf [
  rdf:type      owl:Restriction ;
  owl:onProperty :hasPrimaryPhysician ;
  owl:maxCardinality 1
] .
```

OWL (Web Ontology Language)

- ▶ Using Turtle syntax, write RDF triples to represent:
 - ▶ Lion is a type of Animal
 - ▶ `:Lion rdf:type :Animal .`
 - ▶ Lion lives in Africa
 - ▶ `:Lion :livesIn :Africa .`
 - ▶ Africa is a type of Continent
 - ▶ `:Africa rdf:type :Continent .`

OWL (Web Ontology Language)

- ▶ Declare Animal and Continent as classes
- ▶ Constrain *livesIn* so its domain is Animal and its range is Continent

```
:Animal
  rdf:type owl:Class ;
  rdfs:label "Animal"@en .
:Continent
  rdf:type owl:Class ;
  rdfs:label "Continent"@en .
# --- Property declaration with domain and range
:livesIn
  rdf:type rdf:Property ;
  rdfs:domain :Animal ;
  rdfs:range :Continent ;
  rdfs:label "lives in"@en .
:Lion rdf:type :Animal .
:Africa rdf:type :Continent .
:Lion :livesIn :Africa .
```

- ▶ Friend of Friend = Friend
- ▶ Enemy of Enemy = Friend
- ▶ Enemy of Friend = Enemy
- ▶ Friend of Enemy = Enemy
- ▶ Jim is my Enemy
- ▶ But it turns out that jim is also his own worst enemy